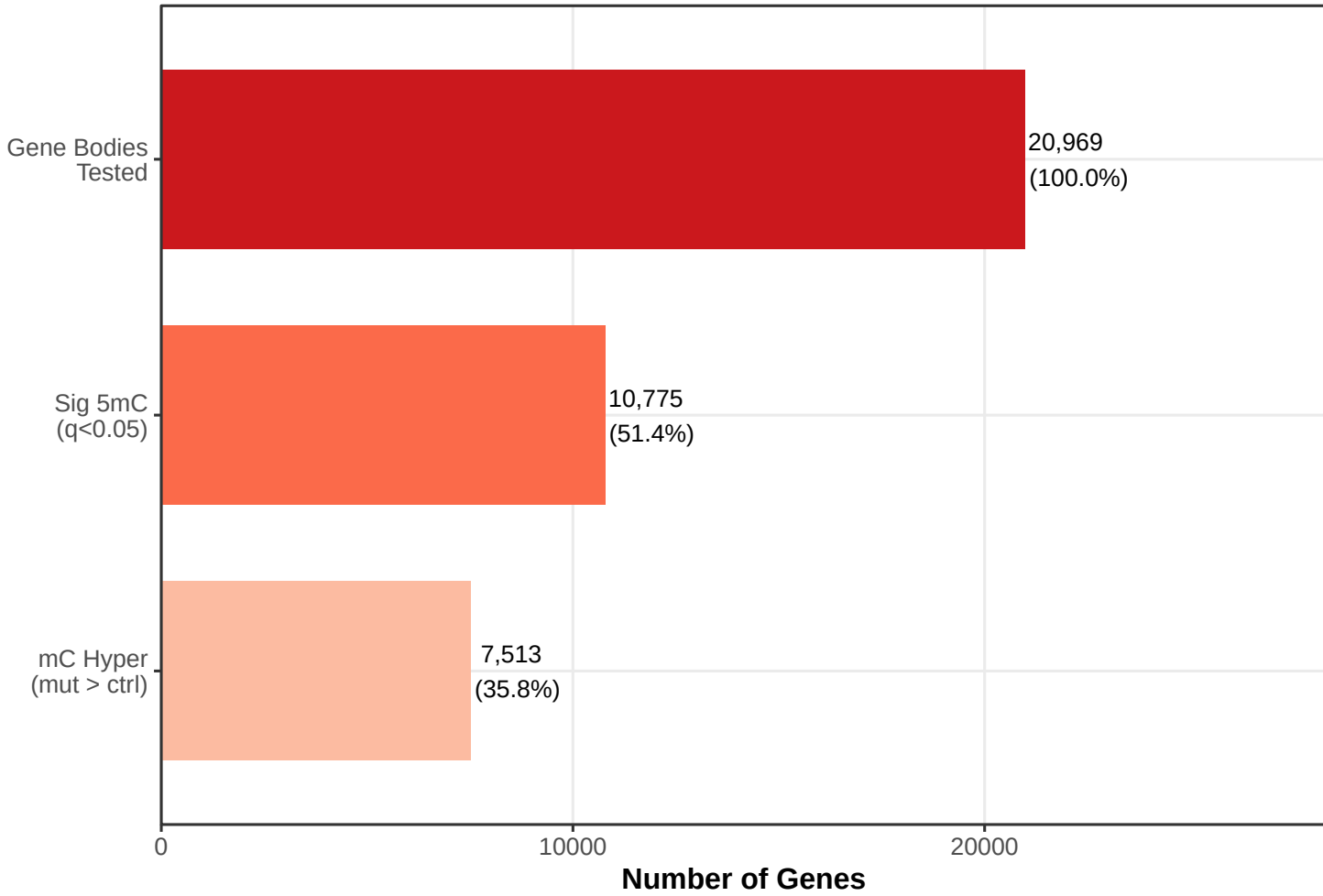


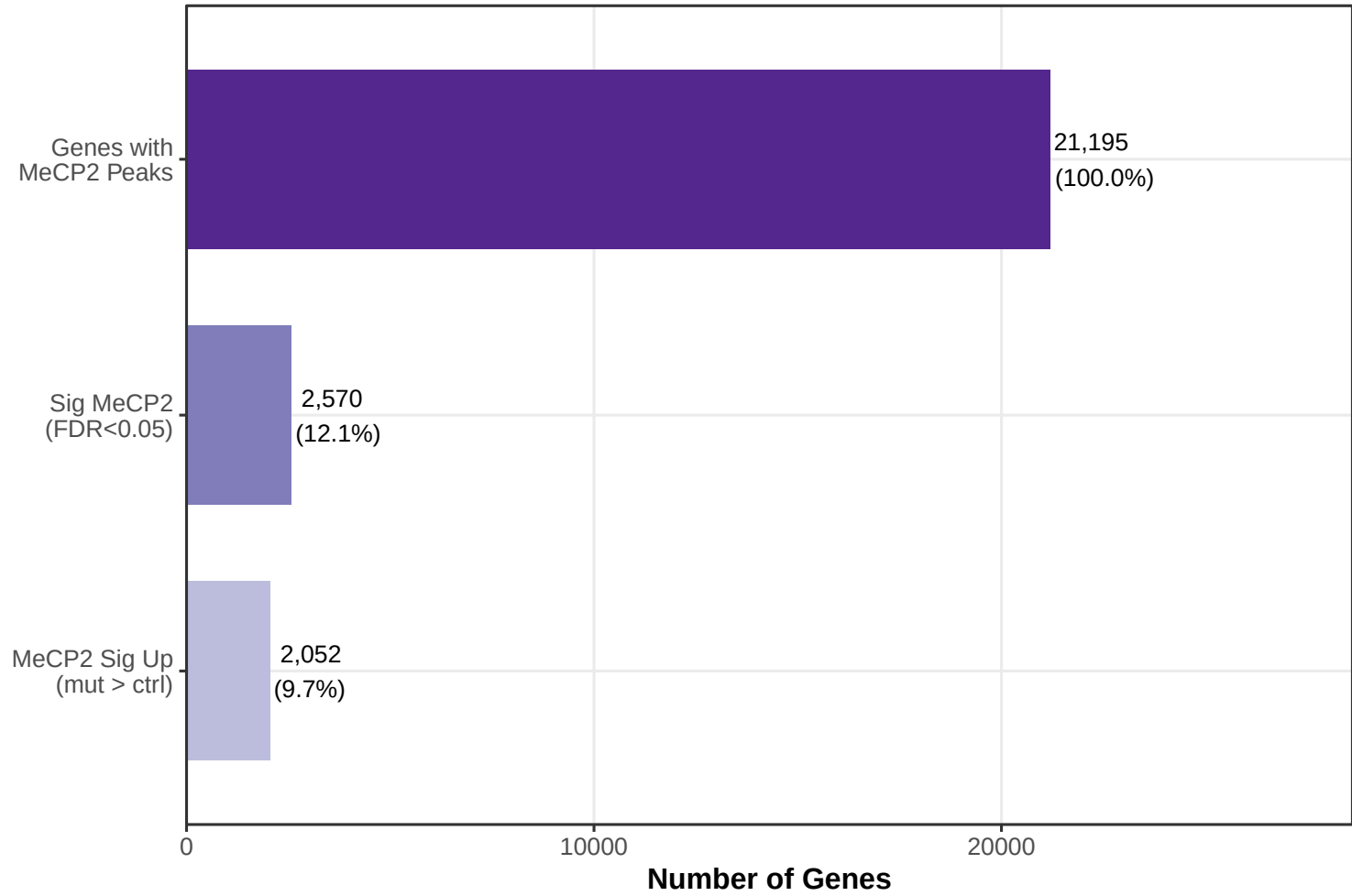
MeCP2 is a Context-Dependent Methylation Reader

Far more genes show methylation changes than MeCP2 binding changes – methylation is necessary but not sufficient

Methylation (5mC) Hierarchy

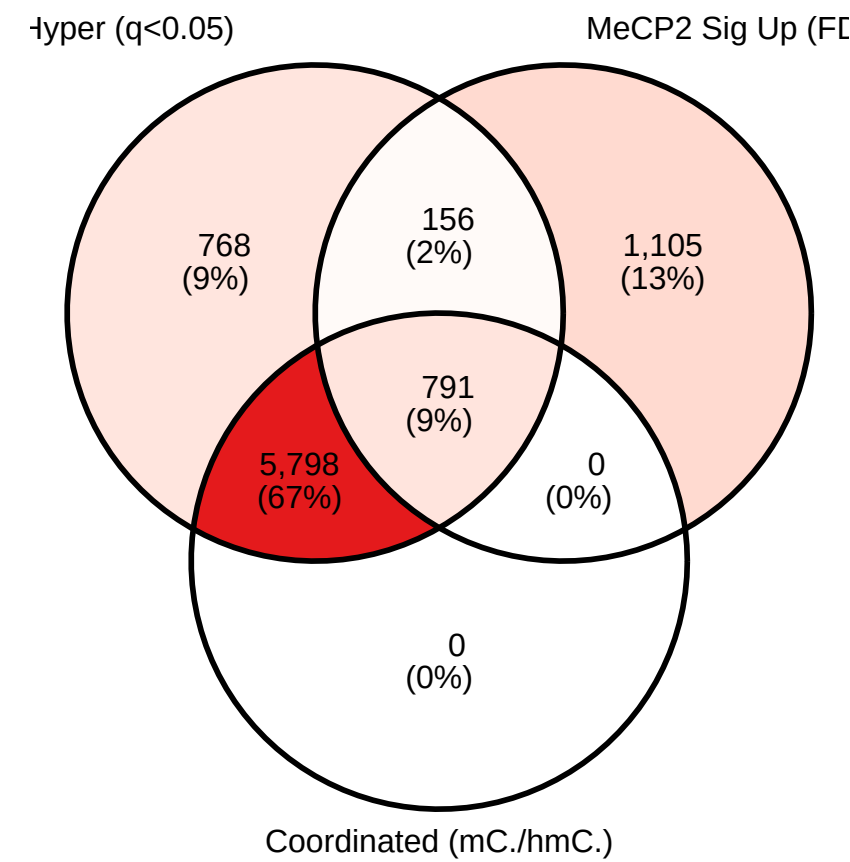


MeCP2 Binding Hierarchy



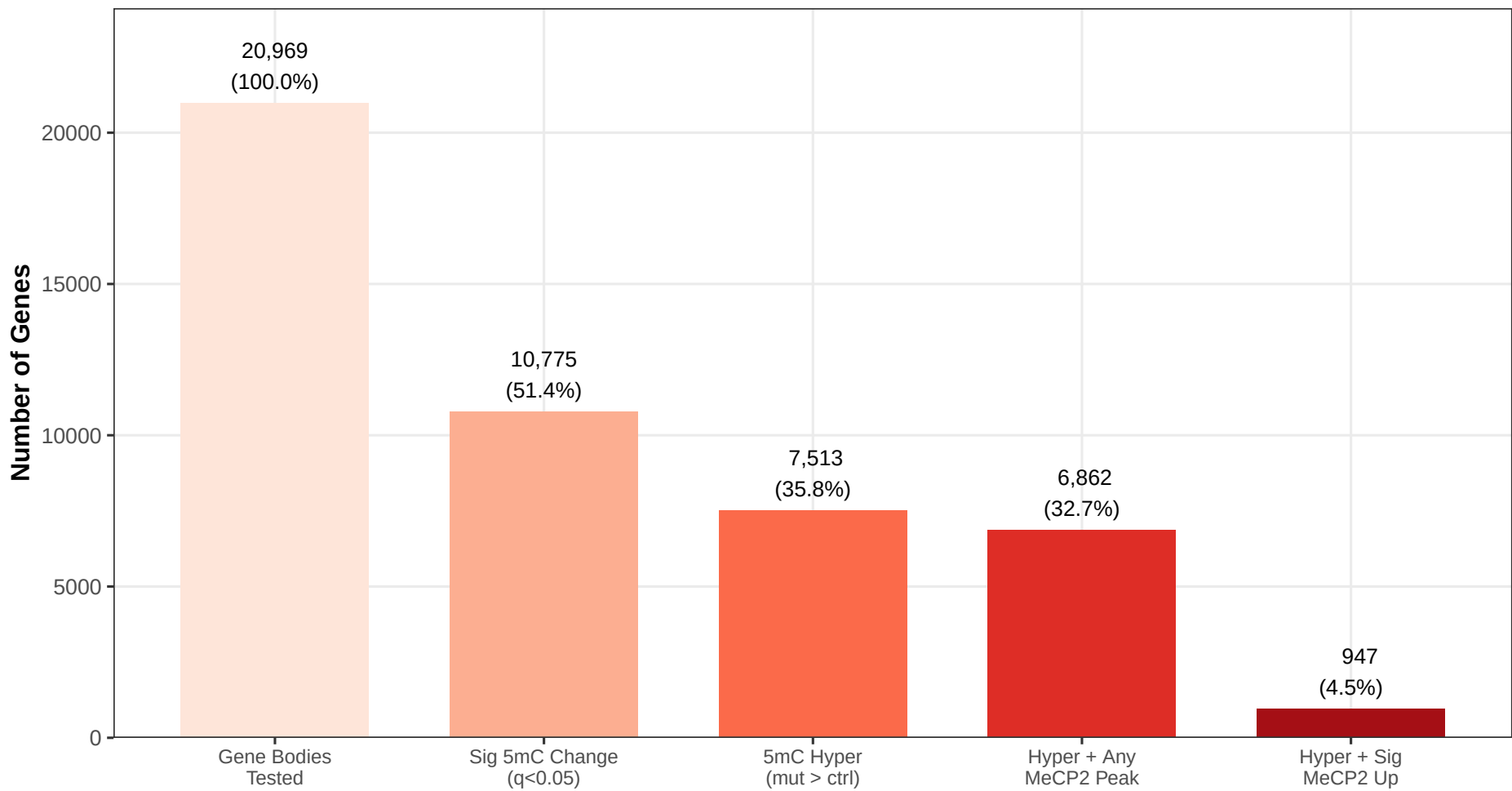
Gene-Level Overlap: Methylation vs MeCP2 Binding

Most mC hyper genes do NOT show significant MeCP2 binding increase



Methylation-to-MeCP2 Binding Funnel

Most methylated genes are NOT bound by MeCP2 – context-dependent recruitment



MeCP2 ChIP Signal at Gene Bodies by Methylation Status

Flipped perspective: how much MeCP2 binds at methylated vs unmethylated CpGs

